

Inversion of Matrices

Outline

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Properties of the Inverse Matrix

Introduction: Properties of the Inverse Matrix

- ▶ Before computing inverse matrices, we collect their main algebraic properties.
- ▶ We show that the inverse is unique, and how it behaves with respect to products, transposition and determinants.
- ▶ These properties are used constantly in linear algebra, e.g., to solve linear systems $\mathbf{Ax} = \mathbf{b}$ as $\mathbf{x} = \mathbf{A}^{-1}\mathbf{b}$.

Invertible and Singular Matrices

- ▶ Recall (see the chapter on matrices) that a square matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$ is **invertible** if there exists a matrix $\mathbf{A}^{-1} \in \mathbb{R}^{n \times n}$ such that

$$\mathbf{A}\mathbf{A}^{-1} = \mathbf{A}^{-1}\mathbf{A} = \mathbf{I}.$$

- ▶ A square matrix which is not invertible is called **singular**.
- ▶ In this chapter, $\mathbf{A}'$ denotes the transpose of $\mathbf{A}$, as in the chapter on matrices.

Proposition 1: Uniqueness of the inverse

Let $\mathbf{A} \in \mathbb{R}^{n \times n}$ be invertible. If $\mathbf{B}, \mathbf{C} \in \mathbb{R}^{n \times n}$ are such that $\mathbf{AB} = \mathbf{BA} = \mathbf{I}$ and $\mathbf{AC} = \mathbf{CA} = \mathbf{I}$, then $\mathbf{B} = \mathbf{C}$.

Proof

Using the associativity of the matrix product, we have:

$$\begin{aligned} \mathbf{B} &= \mathbf{B}\mathbf{I} \\ &= \mathbf{B}(\mathbf{A}\mathbf{C}) \\ &= (\mathbf{B}\mathbf{A})\mathbf{C} \\ &= \mathbf{I}\mathbf{C} = \mathbf{C}. \end{aligned}$$



Proposition 2: Properties of the inverse

Let $\mathbf{A}, \mathbf{B} \in \mathbb{R}^{n \times n}$ be invertible matrices. Then:

1. $\mathbf{A}^{-1}$ is invertible and $(\mathbf{A}^{-1})^{-1} = \mathbf{A}$;
2. $\mathbf{AB}$ is invertible and $(\mathbf{AB})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1}$;
3. $\mathbf{A}'$ is invertible and $(\mathbf{A}')^{-1} = (\mathbf{A}^{-1})'$;
4. $\det(\mathbf{A}) \neq 0$ and $\det(\mathbf{A}^{-1}) = \frac{1}{\det(\mathbf{A})}$.

Proof (1/3)

By the uniqueness of the inverse, in each case it is enough to exhibit a matrix that multiplied on the left and on the right gives I .

1. From $A^{-1}A = AA^{-1} = I$, the matrix A is the inverse of A^{-1} .
2. Using associativity:

$$(AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1} = AIA^{-1} = AA^{-1} = I,$$

$$(B^{-1}A^{-1})(AB) = B^{-1}(A^{-1}A)B = B^{-1}IB = B^{-1}B = I.$$

Proof (2/3)

3. Recall that $(\mathbf{CD})' = \mathbf{D}'\mathbf{C}'$ for all $\mathbf{C}, \mathbf{D} \in \mathbb{R}^{n \times n}$: indeed, the entry (i, j) of $(\mathbf{CD})'$ is the entry (j, i) of $\mathbf{CD}$, i.e.,

$$\sum_{k=1}^n c_{jk} d_{ki} = \sum_{k=1}^n d'_{ik} c'_{kj},$$

which is the entry (i, j) of $\mathbf{D}'\mathbf{C}'$. Then, since $\mathbf{I}' = \mathbf{I}$:

$$\mathbf{A}'(\mathbf{A}^{-1})' = (\mathbf{A}^{-1}\mathbf{A})' = \mathbf{I}' = \mathbf{I},$$

$$(\mathbf{A}^{-1})'\mathbf{A}' = (\mathbf{A}\mathbf{A}^{-1})' = \mathbf{I}' = \mathbf{I}.$$

Proof (3/3)

4. From the product rule of determinants:

$$\det(\mathbf{A}) \det(\mathbf{A}^{-1}) = \det(\mathbf{A}\mathbf{A}^{-1}) = \det(\mathbf{I}) = 1.$$

Hence

$$\det(\mathbf{A}) \neq 0 \quad \text{and} \quad \det(\mathbf{A}^{-1}) = \frac{1}{\det(\mathbf{A})}.$$



Remarks

- ▶ Note the reversed order in $(\mathbf{AB})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1}$: since the matrix product is not commutative, in general

$$(\mathbf{AB})^{-1} \neq \mathbf{A}^{-1}\mathbf{B}^{-1}.$$

- ▶ Property 4 shows that a matrix with $\det(\mathbf{A}) = 0$ cannot be invertible, i.e., it is singular.

Summary

Inverse matrix:

- ▶ $\mathbf{A}\mathbf{A}^{-1} = \mathbf{A}^{-1}\mathbf{A} = \mathbf{I}$; a non-invertible matrix is singular
- ▶ The inverse, if it exists, is unique

Properties:

- ▶ $(\mathbf{A}^{-1})^{-1} = \mathbf{A}$, $(\mathbf{AB})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1}$ (reversed order)
- ▶ $(\mathbf{A}')^{-1} = (\mathbf{A}^{-1})'$
- ▶ $\det(\mathbf{A}^{-1}) = 1/\det(\mathbf{A})$: if $\det(\mathbf{A}) = 0$, then $\mathbf{A}$ is singular

Computation of the Inverse

Introduction: Computation of the Inverse

- ▶ We present two methods to compute the inverse of a matrix:
 - ▶ the **Gauss–Jordan method**, based on elementary row operations;
 - ▶ the **adjugate formula**, based on cofactors and determinants.
- ▶ We also see how to recognize a singular matrix during the computation.
- ▶ Finally, we prove that $\mathbf{A}$ is invertible if and only if $\det(\mathbf{A}) \neq 0$.

Gauss–Jordan Method

Gauss–Jordan method (idea).

To compute $\mathbf{A}^{-1}$, we build the augmented matrix

$$(\mathbf{A} \mid \mathbf{I})$$

and we apply **elementary row operations** to transform the left block into the identity matrix.

If we obtain

$$(\mathbf{I} \mid \mathbf{B}),$$

then $\mathbf{B} = \mathbf{A}^{-1}$.

Example (1/4)

Let us consider

$$\mathbf{A} = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}.$$

We compute the inverse matrix $\mathbf{A}^{-1}$ with the method of Gauss–Jordan.

We form the augmented matrix $(\mathbf{A} \mid \mathbf{I})$:

$$\left(\begin{array}{cc|cc} 2 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 \end{array} \right)$$

Example (2/4)

Step 1: Swap rows 1 and 2 to get a 1 in the pivot position.

$$\left(\begin{array}{cc|cc} 1 & 1 & 0 & 1 \\ 2 & 1 & 1 & 0 \end{array} \right) \quad (R_1 \leftrightarrow R_2)$$

Step 2: Eliminate below the first pivot.

$$\left(\begin{array}{cc|cc} 1 & 1 & 0 & 1 \\ 0 & -1 & 1 & -2 \end{array} \right) \quad (R_2 \leftarrow R_2 - 2R_1)$$

Example (3/4)

Step 3: Scale row 2 to get a 1 in the second pivot position.

$$\left(\begin{array}{cc|cc} 1 & 1 & 0 & 1 \\ 0 & 1 & -1 & 2 \end{array} \right) \quad (R_2 \leftarrow -R_2)$$

Step 4: Eliminate above the second pivot.

$$\left(\begin{array}{cc|cc} 1 & 0 & 1 & -1 \\ 0 & 1 & -1 & 2 \end{array} \right) \quad (R_1 \leftarrow R_1 - R_2)$$

Example (4/4)

Since the left block is the identity matrix, we obtain:

$$\mathbf{A}^{-1} = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix}.$$

We can verify:

$$\mathbf{A}\mathbf{A}^{-1} = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \mathbf{I}.$$

Example (1/5)

Let us consider

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 2 & 0 & 1 \end{pmatrix}.$$

We compute the inverse matrix $\mathbf{A}^{-1}$ with the method of Gauss–Jordan.

We form the augmented matrix $(\mathbf{A} \mid \mathbf{I})$:

$$\left(\begin{array}{ccc|ccc} 1 & 2 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 2 & 0 & 1 & 0 & 0 & 1 \end{array} \right)$$

Example (2/5)

Step 1: Eliminate below the first pivot.

$$\left(\begin{array}{ccc|ccc} 1 & 2 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & -4 & 1 & -2 & 0 & 1 \end{array} \right) \quad (R_3 \leftarrow R_3 - 2R_1)$$

Step 2: Eliminate above the second pivot (in row 1).

$$\left(\begin{array}{ccc|ccc} 1 & 0 & -2 & 1 & -2 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & -4 & 1 & -2 & 0 & 1 \end{array} \right) \quad (R_1 \leftarrow R_1 - 2R_2)$$

Example (3/5)

Step 3: Eliminate below the second pivot (in row 3).

$$\left(\begin{array}{ccc|ccc} 1 & 0 & -2 & 1 & -2 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 5 & -2 & 4 & 1 \end{array} \right) \quad (R_3 \leftarrow R_3 + 4R_2)$$

Step 4: Scale row 3 to get a 1 in the third pivot position.

$$\left(\begin{array}{ccc|ccc} 1 & 0 & -2 & 1 & -2 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & -\frac{2}{5} & \frac{4}{5} & \frac{1}{5} \end{array} \right) \quad (R_3 \leftarrow \frac{1}{5}R_3)$$

Example (4/5)

Step 5: Eliminate above the third pivot (in row 1).

$$\left(\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{1}{5} & -\frac{2}{5} & \frac{2}{5} \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & -\frac{2}{5} & \frac{4}{5} & \frac{1}{5} \end{array} \right) \quad (R_1 \leftarrow R_1 + 2R_3)$$

Step 6: Eliminate above the third pivot (in row 2).

$$\left(\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{1}{5} & -\frac{2}{5} & \frac{2}{5} \\ 0 & 1 & 0 & \frac{2}{5} & \frac{1}{5} & -\frac{1}{5} \\ 0 & 0 & 1 & -\frac{2}{5} & \frac{4}{5} & \frac{1}{5} \end{array} \right) \quad (R_2 \leftarrow R_2 - R_3)$$

Example (5/5)

Since the left block is the identity matrix, we obtain:

$$\mathbf{A}^{-1} = \begin{pmatrix} \frac{1}{5} & -\frac{2}{5} & \frac{2}{5} \\ \frac{2}{5} & \frac{1}{5} & -\frac{1}{5} \\ -\frac{2}{5} & \frac{4}{5} & \frac{1}{5} \end{pmatrix} = \frac{1}{5} \begin{pmatrix} 1 & -2 & 2 \\ 2 & 1 & -1 \\ -2 & 4 & 1 \end{pmatrix}.$$

Example (1/5)

Let us consider

$$\mathbf{A} = \begin{pmatrix} 0 & 1 & 2 \\ 1 & 0 & 3 \\ 4 & -3 & 8 \end{pmatrix}.$$

We compute the inverse matrix $\mathbf{A}^{-1}$ with the method of Gauss–Jordan. Since $a_{11} = 0$, we start with a row swap.

We form the augmented matrix $(\mathbf{A} \mid \mathbf{I})$:

$$\left(\begin{array}{ccc|ccc} 0 & 1 & 2 & 1 & 0 & 0 \\ 1 & 0 & 3 & 0 & 1 & 0 \\ 4 & -3 & 8 & 0 & 0 & 1 \end{array} \right)$$

Example (2/5)

$$\left(\begin{array}{ccc|ccc} 1 & 0 & 3 & 0 & 1 & 0 \\ 0 & 1 & 2 & 1 & 0 & 0 \\ 4 & -3 & 8 & 0 & 0 & 1 \end{array} \right) \quad (R_1 \leftrightarrow R_2)$$

$$\left(\begin{array}{ccc|ccc} 1 & 0 & 3 & 0 & 1 & 0 \\ 0 & 1 & 2 & 1 & 0 & 0 \\ 0 & -3 & -4 & 0 & -4 & 1 \end{array} \right) \quad (R_3 \leftarrow R_3 - 4R_1)$$

Example (3/5)

$$\left(\begin{array}{ccc|ccc} 1 & 0 & 3 & 0 & 1 & 0 \\ 0 & 1 & 2 & 1 & 0 & 0 \\ 0 & 0 & 2 & 3 & -4 & 1 \end{array} \right) \quad (R_3 \leftarrow R_3 + 3R_2)$$

$$\left(\begin{array}{ccc|ccc} 1 & 0 & 3 & 0 & 1 & 0 \\ 0 & 1 & 2 & 1 & 0 & 0 \\ 0 & 0 & 1 & \frac{3}{2} & -2 & \frac{1}{2} \end{array} \right) \quad (R_3 \leftarrow \frac{1}{2}R_3)$$

Example (4/5)

$$\left(\begin{array}{ccc|ccc} 1 & 0 & 0 & -\frac{9}{2} & 7 & -\frac{3}{2} \\ 0 & 1 & 2 & 1 & 0 & 0 \\ 0 & 0 & 1 & \frac{3}{2} & -2 & \frac{1}{2} \end{array} \right) \quad (R_1 \leftarrow R_1 - 3R_3)$$

$$\left(\begin{array}{ccc|ccc} 1 & 0 & 0 & -\frac{9}{2} & 7 & -\frac{3}{2} \\ 0 & 1 & 0 & -2 & 4 & -1 \\ 0 & 0 & 1 & \frac{3}{2} & -2 & \frac{1}{2} \end{array} \right) \quad (R_2 \leftarrow R_2 - 2R_3)$$

Example (5/5)

Since the left block is the identity matrix, we obtain:

$$\mathbf{A}^{-1} = \begin{pmatrix} -\frac{9}{2} & 7 & -\frac{3}{2} \\ -2 & 4 & -1 \\ \frac{3}{2} & -2 & \frac{1}{2} \end{pmatrix} = \frac{1}{2} \begin{pmatrix} -9 & 14 & -3 \\ -4 & 8 & -2 \\ 3 & -4 & 1 \end{pmatrix}.$$

What If the Matrix Is Singular?

The Gauss–Jordan method also tells us
when the inverse does **not** exist.

Observation 1: Singular matrices and the Gauss–Jordan method

Let $\mathbf{A} \in \mathbb{R}^{n \times n}$. If, while applying elementary row operations to $(\mathbf{A} \mid \mathbf{I})$, a zero row appears in the left block, then $\det(\mathbf{A}) = 0$ and $\mathbf{A}$ is singular: the method stops and $\mathbf{A}^{-1}$ does not exist.

Proof

- ▶ The left block is obtained from $\mathbf{A}$ by elementary row operations.
- ▶ Each of them multiplies the determinant by a nonzero number (-1 for a row swap, $\lambda \neq 0$ for a row scaling, 1 for a row addition).
- ▶ A matrix with a zero row has zero determinant (Laplace expansion along that row), hence $\det(\mathbf{A}) = 0$.
- ▶ By property 4 of Proposition 2, $\mathbf{A}$ is not invertible.



Example (1/3)

Let us consider

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 5 & 7 \\ 1 & 3 & 4 \end{pmatrix}.$$

We try to compute $\mathbf{A}^{-1}$ with the method of Gauss–Jordan.

$$\left(\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 2 & 5 & 7 & 0 & 1 & 0 \\ 1 & 3 & 4 & 0 & 0 & 1 \end{array} \right)$$

Example (2/3)

$$\left(\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & 1 & -2 & 1 & 0 \\ 1 & 3 & 4 & 0 & 0 & 1 \end{array} \right) \quad (R_2 \leftarrow R_2 - 2R_1)$$

$$\left(\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & 1 & -2 & 1 & 0 \\ 0 & 1 & 1 & -1 & 0 & 1 \end{array} \right) \quad (R_3 \leftarrow R_3 - R_1)$$

$$\left(\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & 1 & -2 & 1 & 0 \\ 0 & 0 & 0 & 1 & -1 & 1 \end{array} \right) \quad (R_3 \leftarrow R_3 - R_2)$$

Example (3/3)

- ▶ The third row of the left block is zero: the left block cannot be transformed into the identity matrix, and the method stops.
- ▶ Hence $\mathbf{A}$ is singular.
- ▶ Indeed, the third row of $\mathbf{A}$ is the difference of the second and the first row, and

$$\begin{aligned}\det(\mathbf{A}) &= 1 \cdot (5 \cdot 4 - 7 \cdot 3) - 2 \cdot (2 \cdot 4 - 7 \cdot 1) \\ &\quad + 3 \cdot (2 \cdot 3 - 5 \cdot 1) \\ &= -1 - 2 + 3 = 0.\end{aligned}$$

Inverse via the Adjugate Matrix

We now derive an explicit formula for the inverse,
based on cofactors and determinants.

Cofactors

- Recall (see the chapter on matrices) that, given $\mathbf{A} \in \mathbb{R}^{n \times n}$, the **cofactor** of the entry a_{ij} is

$$C_{ij} = (-1)^{i+j} \det(\mathbf{A}_{ij}),$$

where $\mathbf{A}_{ij}$ is the minor matrix obtained from $\mathbf{A}$ by deleting row i and column j .

Definition 1: Cofactor matrix and adjugate matrix

Let $\mathbf{A} \in \mathbb{R}^{n \times n}$ with $n \geq 2$. The **cofactor matrix** of $\mathbf{A}$ is the matrix $\mathbf{C} = (C_{ij}) \in \mathbb{R}^{n \times n}$ of the cofactors of $\mathbf{A}$. The **adjugate matrix** of $\mathbf{A}$ is the transpose of the cofactor matrix:

$$\text{adj}(\mathbf{A}) = \mathbf{C}', \quad \text{i.e.,} \quad (\text{adj}(\mathbf{A}))_{ij} = C_{ji}.$$

Theorem 1: Adjugate formula

Let $\mathbf{A} \in \mathbb{R}^{n \times n}$ with $n \geq 2$. Then

$$\mathbf{A} \operatorname{adj}(\mathbf{A}) = \operatorname{adj}(\mathbf{A}) \mathbf{A} = \det(\mathbf{A}) \mathbf{I}.$$

Consequently, if $\det(\mathbf{A}) \neq 0$, then $\mathbf{A}$ is invertible and

$$\mathbf{A}^{-1} = \frac{1}{\det(\mathbf{A})} \operatorname{adj}(\mathbf{A}).$$

Proof (1/2)

The entry (i, j) of $\mathbf{A} \operatorname{adj}(\mathbf{A})$ is

$$\sum_{k=1}^n a_{ik} C_{jk}.$$

- ▶ If $i = j$, this is the Laplace expansion of $\det(\mathbf{A})$ along row i .
- ▶ If $i \neq j$, this is the Laplace expansion along row j of the matrix $\tilde{\mathbf{A}}$ obtained from $\mathbf{A}$ by replacing row j with row i (the cofactors C_{jk} do not depend on row j).
- ▶ The matrix $\tilde{\mathbf{A}}$ has two equal rows: swapping them leaves $\tilde{\mathbf{A}}$ unchanged and changes the sign of its determinant, so $\det(\tilde{\mathbf{A}}) = -\det(\tilde{\mathbf{A}})$, i.e., $\det(\tilde{\mathbf{A}}) = 0$.

Proof (2/2)

- ▶ Hence

$$\mathbf{A} \operatorname{adj}(\mathbf{A}) = \det(\mathbf{A}) \mathbf{I}.$$

- ▶ The identity $\operatorname{adj}(\mathbf{A}) \mathbf{A} = \det(\mathbf{A}) \mathbf{I}$ is obtained in the same way, using Laplace expansions along columns.
- ▶ If $\det(\mathbf{A}) \neq 0$, dividing by $\det(\mathbf{A})$ shows that $\frac{1}{\det(\mathbf{A})} \operatorname{adj}(\mathbf{A})$ is the inverse of $\mathbf{A}$.



Observation 2: Invertibility and determinant

A square matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$ is invertible if and only if $\det(\mathbf{A}) \neq 0$.

Proof

- ▶ If $\mathbf{A}$ is invertible, then $\det(\mathbf{A}) \neq 0$ by property 4 of Proposition 2.
- ▶ Conversely, if $\det(\mathbf{A}) \neq 0$, then $\mathbf{A}$ is invertible by Theorem 1.
- ▶ For $n = 1$, $\mathbf{A} = (a)$ with $a \neq 0$ and $\mathbf{A}^{-1} = (1/a)$.



Observation 3: Inverse of a 2×2 matrix

Let

$$\mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathbb{R}^{2 \times 2} \quad \text{with} \quad \det(\mathbf{A}) = ad - bc \neq 0.$$

Then

$$\mathbf{A}^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

Proof

The minor matrices of $\mathbf{A}$ are 1×1 , so the cofactors are

$$C_{11} = d, \quad C_{12} = -c, \quad C_{21} = -b, \quad C_{22} = a.$$

Hence

$$\mathbf{C} = \begin{pmatrix} d & -c \\ -b & a \end{pmatrix}, \quad \text{adj}(\mathbf{A}) = \mathbf{C}' = \begin{pmatrix} d & -b \\ -c & a \end{pmatrix},$$

and the formula follows from Theorem 1. □

Inverse of a 2×2 Matrix (Rule)

Swap the two diagonal entries,
change the sign of the two off-diagonal entries,
and divide by the determinant.

Example

Let us consider again

$$\mathbf{A} = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}.$$

We have $\det(\mathbf{A}) = 2 \cdot 1 - 1 \cdot 1 = 1 \neq 0$, hence

$$\mathbf{A}^{-1} = \frac{1}{1} \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix} = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix},$$

which is the matrix obtained with the method of Gauss–Jordan in the first example.

Example (1/4)

Let us consider again

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 2 & 0 & 1 \end{pmatrix}.$$

By Laplace expansion along the first row:

$$\det(\mathbf{A}) = 1 \cdot (1 \cdot 1 - 1 \cdot 0) - 2 \cdot (0 \cdot 1 - 1 \cdot 2) + 0 = 1 + 4 = 5 \neq 0.$$

Example (2/4)

The cofactors are:

$$C_{11} = + \det \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = 1, \quad C_{21} = - \det \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix} = -2,$$

$$C_{12} = - \det \begin{pmatrix} 0 & 1 \\ 2 & 1 \end{pmatrix} = 2, \quad C_{22} = + \det \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix} = 1,$$

$$C_{13} = + \det \begin{pmatrix} 0 & 1 \\ 2 & 0 \end{pmatrix} = -2, \quad C_{23} = - \det \begin{pmatrix} 1 & 2 \\ 2 & 0 \end{pmatrix} = 4,$$

$$C_{31} = + \det \begin{pmatrix} 2 & 0 \\ 1 & 1 \end{pmatrix} = 2, \quad C_{32} = - \det \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = -1,$$

$$C_{33} = + \det \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix} = 1.$$

Example (3/4)

Hence

$$\mathbf{C} = \begin{pmatrix} 1 & 2 & -2 \\ -2 & 1 & 4 \\ 2 & -1 & 1 \end{pmatrix}, \quad \text{adj}(\mathbf{A}) = \mathbf{C}' = \begin{pmatrix} 1 & -2 & 2 \\ 2 & 1 & -1 \\ -2 & 4 & 1 \end{pmatrix}.$$

Example (4/4)

Therefore

$$\mathbf{A}^{-1} = \frac{1}{\det(\mathbf{A})} \operatorname{adj}(\mathbf{A}) = \frac{1}{5} \begin{pmatrix} 1 & -2 & 2 \\ 2 & 1 & -1 \\ -2 & 4 & 1 \end{pmatrix},$$

which is the matrix obtained with the method of Gauss–Jordan in the second example.

Which Method to Use?

- ▶ For $n = 2$ (and often for $n = 3$) the adjugate formula is convenient for computations by hand.
- ▶ For larger matrices, the method of Gauss–Jordan requires far fewer operations.

Summary (1/2)

Gauss–Jordan method:

- ▶ Form the augmented matrix $(\mathbf{A} \mid \mathbf{I})$
- ▶ Apply elementary row operations (swaps, scalings, additions) until $(\mathbf{I} \mid \mathbf{B})$: then $\mathbf{B} = \mathbf{A}^{-1}$
- ▶ A row swap is needed when the pivot is zero

Singular matrices:

- ▶ A zero row in the left block $\Rightarrow \det(\mathbf{A}) = 0$: the method stops, $\mathbf{A}^{-1}$ does not exist

Summary (2/2)

Adjugate formula:

- ▶ $\text{adj}(\mathbf{A}) = \mathbf{C}'$, with $C_{ij} = (-1)^{i+j} \det(\mathbf{A}_{ij})$
- ▶ $\mathbf{A} \text{adj}(\mathbf{A}) = \text{adj}(\mathbf{A}) \mathbf{A} = \det(\mathbf{A}) \mathbf{I}$
- ▶ $\mathbf{A}^{-1} = \frac{1}{\det(\mathbf{A})} \text{adj}(\mathbf{A})$; for 2×2 : swap diagonal, change sign off-diagonal, divide by $ad - bc$

Invertibility:

- ▶ $\mathbf{A}$ is invertible if and only if $\det(\mathbf{A}) \neq 0$

Key Takeaways

Summary

1. Properties of the inverse:

- ▶ Unique; $(\mathbf{AB})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1}$, $(\mathbf{A}')^{-1} = (\mathbf{A}^{-1})'$,
 $\det(\mathbf{A}^{-1}) = 1/\det(\mathbf{A})$

2. Gauss–Jordan method:

- ▶ $(\mathbf{A} \mid \mathbf{I}) \rightarrow (\mathbf{I} \mid \mathbf{A}^{-1})$ by elementary row operations
- ▶ A zero row in the left block means that $\mathbf{A}$ is singular

3. Adjugate formula:

- ▶ $\mathbf{A}^{-1} = \frac{1}{\det(\mathbf{A})} \text{adj}(\mathbf{A})$, convenient by hand for $n \leq 3$
- ▶ $\mathbf{A}$ invertible $\iff \det(\mathbf{A}) \neq 0$